GLOBAL RESERVE TOKEN

How GRETVault works

A plain-language guide to the vault contract: how GRET is minted and redeemed, how the basket changes, what governance and the rebalancer can and cannot do, and what happens in an emergency.

November 2, 2026

Covers GRETVault with live-balance accounting, ChainlinkPriceGuard, and every accepted risk

Not yet deployed · for a mainnet pilot with the project's own funds · not independently audited

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CHAPTER 1

The big picture

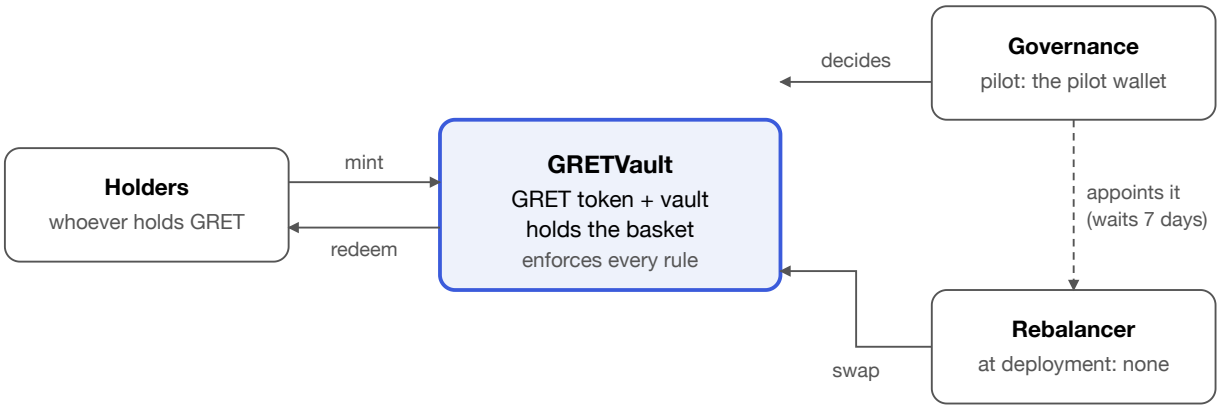
GRET is a share of a basket of tokenized assets. One contract issues the shares and holds the basket, and it enforces every rule on its own.

1.1 One contract, one token

`GRETVault` is two things at once. It is the GRET token, a standard ERC-20 that wallets such as MetaMask can show and that can be transferred freely. And it is the vault that holds the basket: 11 tokens, from ITOTon and SPYx to PAXG and cbBTC. The token's address is the vault's address.

Holding GRET means owning a proportional part of everything the vault holds of its active tokens. If 100 GRET exist, each one represents 1% of each of those balances. Minting and redeeming use no prices and no stored value per GRET: the contract only counts tokens, reading what it holds at that moment. Prices appear in one place only, to limit what a swap can lose (chapter 7).

1.2 Who does what

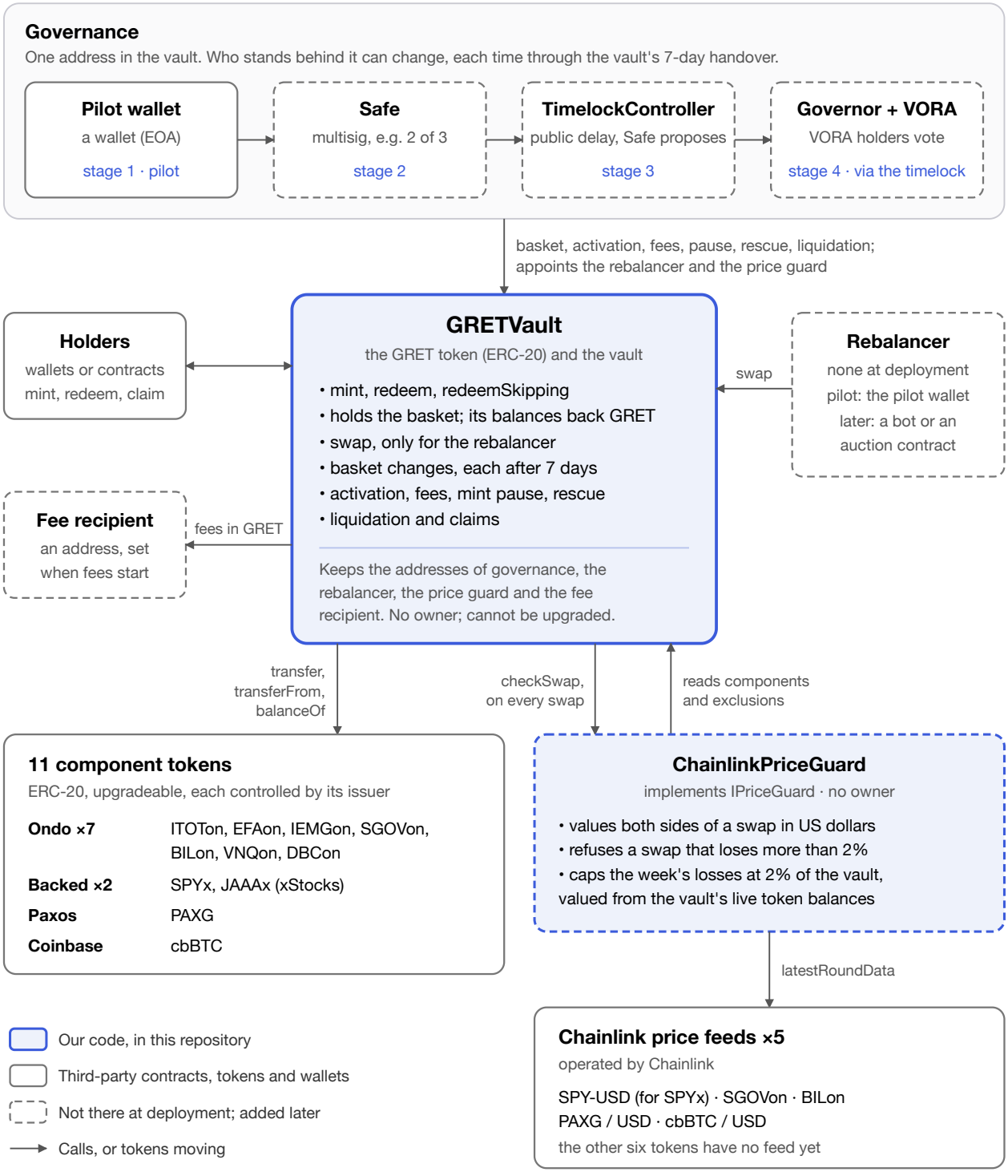


Holders enter and exit through the vault. Governance decides changes and appoints the rebalancer, the only address that can swap one token for another, within the limits of a price guard.

Role	What it does	In the pilot
Holders	Mint GRET by depositing the basket and redeem it for the basket. Anyone can hold GRET; only wallets that minted can redeem it normally.	—
Governance	Changes the basket, activates and excludes tokens, sets fees, pauses minting, appoints the rebalancer, the price guard and its own successor, rescues tokens outside the basket, and can start a liquidation. Every change that could take value waits 7 days.	The pilot wallet (0xe03E… b7fd)

Rebalancer	Exchanges one basket token for another (a swap). It is the only role that moves the basket at a price it chooses.	None at deployment
Price guard	A contract with no owner that prices every swap with Chainlink and refuses the ones that lose too much.	None at deployment
Fee recipient	Receives the mint and redemption fees, paid in GRET.	Fees are 0
Issuers	Ondo, Backed, Paxos and Coinbase control their own tokens: they can pause, freeze or upgrade them. The vault cannot stop that.	—

1.3 The contracts and how they connect



Every contract around the vault. Only GRETVault holds funds. The price guard can only refuse a swap; it never moves tokens. It reads the vault's list of components and which are excluded, and asks each token for the vault's balance. Governance and the rebalancer are just addresses to the vault, so what stands behind them can change without touching the vault.

Contract	What it does	At deployment
In this repository		
`GRETVault`	The GRET token and the vault in one contract. It takes the basket in and issues GRET, pays the basket out for GRET, and backs GRET with its live balance of every active token, with no separate record to keep in step. It runs the rebalancer's swaps, the basket changes with their 7-day waits, component activation, fees, the mint pause, rescue of tokens outside the basket, and liquidation. It enforces every limit on its own. No owner, no upgrades.	Deployed by the pilot wallet, which makes the first mint
`IPriceGuard`	Not a deployed contract but an interface: the one question the vault asks before every swap, `checkSwap(sell, sellAmount, buy, received)`, once the incoming token has arrived and before the outgoing one leaves. Any contract that answers it can be appointed as the guard; if it says no, the swap reverts.	—
`ChainlinkPriceGuard`	The first price guard. It values what the vault gives and what it gets in US dollars with Chainlink feeds, and refuses a swap that loses more than 2%, one that would take the week's losses above 2% of the vault, and one with a token without a fresh price. It values the vault from its live balances, leaving out excluded tokens and the payment that just arrived. It holds no funds and has no owner: its feeds and limits are fixed for good, and only the vault can call it.	Not deployed. When a rebalance is needed: deployed with the five feeds and appointed with the 7-day wait
Governance: one address, which can change (chapter 10)		
Pilot wallet a wallet (EOA), not a contract	Deploys the vault, makes the first mint and is the first governance: one signature, at once. If the pilot needs a rebalance, it also becomes the rebalancer.	Governance
Safe by Safe	A multisig wallet: several signers must approve each transaction, so losing or stealing one key no longer loses or takes governance.	Proposed, stage 2
`TimelockController` by OpenZeppelin	Queues every governance action and runs it only after a public delay, so holders see it coming. It becomes the vault's governance; the Safe is its only proposer and executor.	Proposed, stage 3
`Governor` by OpenZeppelin	Runs VORA votes: proposals, voting, counting, and queueing what passes in the timelock, which stays the vault's governance.	Proposed, stage 4

VORA the project's voting token	Gives voting power in the Governor; it must be built on OpenZeppelin's `ERC20Votes`. It is not in this repository, and GRET itself never votes.	Proposed, stage 4
Other roles, each held by one address		
Rebalancer	The only address that can call `swap`: it delivers one active basket token and takes another, at a price it chooses within the guard's limits. It can be a wallet, or a contract that funds swaps with a flash loan or runs an auction (section 7.5).	None. Appointed with the 7-day wait when needed
Fee recipient	Receives the mint and redemption fees, in GRET. Governance sets it at once; it can never be the vault.	None while fees are zero; in the pilot, the pilot wallet
Third parties the vault depends on		
Component tokens Ondo, Backed, Paxos, Coinbase	The 11 ERC-20 tokens the vault holds. All are upgradeable: their issuers can pause, freeze or change them, and the vault cannot stop that.	The basket of 11
Chainlink price feeds by Chainlink	Publish the US-dollar prices the guard reads: SPY (for SPYx), SGOVon, BILon, PAXG and cbBTC. The guard refuses a price older than 25 hours, so swaps of the stock tokens stop on weekends.	Read only by the guard

The repository also has test-only contracts in `contracts/mocks/` (`MockERC20`, `MockFaultyERC20`, `MockNoReturnERC20`, `MockAggregator`, `MockPriceGuard`). They stand in for tokens, feeds and a guard in the tests and are never deployed.

1.4 Four ideas to keep in mind

1. **The rules live in the vault.** The contract's code cannot change after deployment. Governance decides what to do, but always inside limits the vault enforces: waits, caps and what can never be touched.
2. **The balance is the backing.** What backs GRET is what the vault holds of each active token right now, as the token reports it. Dividends and other income that arrive as balance count at once, and so do losses; nobody has to record them.
3. **Nobody takes the basket for free.** While GRET exists, basket tokens leave the vault only when someone burns GRET (a redemption or a liquidation claim) or when the rebalancer delivers another basket token in exchange.
4. **Big changes are announced.** A new token, a new rebalancer or price guard, a new governance, a write-off or a fee raise waits 7 days, visible on the blockchain, and then must be carried out within the next 7 days or it expires.

CHAPTER 2

Minting GRET

To get new GRET you deposit a little of every active token in the basket, in the same proportions the vault already holds. Anyone can mint.

2.1 How much you deposit

The amount of each token behind 1 GRET is what the vault holds of that token divided by the GRET in circulation. This number is not stored anywhere: on every operation the vault asks each active token for its balance. To mint, you deposit exactly that proportion of every active token.

> **Example**
>
> The vault holds 10 PAXG and 100 GRET are in circulation, so each GRET stands for 0.1 PAXG. To mint 1 GRET you deposit 0.1 PAXG, plus the matching part of each other token. If the vault also holds 0.5 cbBTC, you deposit 0.005 cbBTC too.

A token the vault holds none of (just activated, sold off or lost) is not asked for. Neither is an *excluded* token (just added and not yet activated, or written off; section 5.2), which the vault does not even call. `previewMint(shares)` tells you the exact amounts before you mint.

Because the balances are read live, anything that has already arrived, such as a dividend, is part of what you deposit: you pay for it, so minting after a distribution takes nothing from the holders who were there when it arrived.

2.2 The first mint

When no GRET is in circulation there are no proportions to copy, so the first mint uses the *initial basket*, a fixed list of amounts. It must ask for exactly 1 GRET and it issues exactly 1 GRET. Excluded tokens are left out. In the pilot, the initial basket is worth about USD 100. The same happens again if every holder leaves and the supply returns to zero. Governance can change the initial basket with `setInitialAmounts`, which only matters at zero supply.

Whatever the vault already holds of active tokens at that moment, such as what a last holder left behind with the emergency exit (section 3.4), belongs to that first GRET. While no GRET exists, governance may instead rescue those balances, except written-off tokens (section 5.4).

2.3 Step by step

1. You approve the vault to take each token, then call `mint(shares, maxAmountsIn, minShares, version)`.
2. The vault reads its balance of every active token before taking anything, so everything already there, income and losses alike, counts in full.

3. It computes what to ask for, rounding *up*, and pulls each token from your wallet.
4. It measures what actually arrived and issues GRET for that, rounding *down*.
5. If there is a mint fee, part of the new GRET goes to the fee recipient (chapter 4). The rest goes to you.
6. Your wallet is marked as having minted, which allows it to redeem from then on.

> **Example: a mint with a new token and an xStock**
>
> The vault holds 10 PAXG and 100 SPYx, and 100 GRET are in circulation. cbBTC was just added and activated, and the rebalancer has not bought any yet. You ask for 1 GRET.
>
> 1. The vault asks for 0.1 PAXG and 1 SPYx, and nothing of cbBTC, which it holds none of.
> 2. PAXG arrives exactly. SPYx is an xStock and arrives 2 base units short: 0.999999999999999998 SPYx.
> 3. For each token, the vault works out how much GRET what arrived is worth, and issues the smallest: 0.999999999999999998 GRET, that is, 1 GRET minus 2 wei. You absorb the 2 missing units; nobody else does.

2.4 Protections and limits

- **Rounding favors everyone else.** You pay a hair more and receive a hair less, so minting never lowers the value behind other holders' GRET.
- **You set your own limits.** `maxAmountsIn` caps what you pay for each token, and `minShares` is the least GRET you accept after the fee. If either is crossed, nothing happens. Income that arrives between your quote and your transaction raises what you are asked for; your limits decide whether you still accept.
- **Your limits stay on the right tokens.** `maxAmountsIn` follows the order of `components()`, and `version` is the `basketVersion()` you read with it. If a token is added or removed before your transaction lands, the version no longer matches and the mint reverts, instead of applying your limits to another token. Activating or excluding a token does not change the version: there your limits protect you. A token that starts counting after your quote, with a limit of 0, makes the mint revert.
- **Tokens that round.** xStocks (SPYx, JAAAx) lose 2 or 3 base units per transfer. The vault tolerates up to 10 base units short and simply issues GRET for what arrived. A bigger shortfall, such as a transfer fee, makes the mint revert.
- **When minting is closed.** While governance has paused it (section 6.8), during a liquidation, and if no active token has any balance (the vault refuses to issue GRET backed by nothing).

CHAPTER 3

Redeeming GRET

Redeeming is minting in reverse: you hand in GRET and receive your proportional part of every active token in the basket.

3.1 Who can redeem

Only wallets that have minted at least once can redeem normally. Once a wallet has minted, it can also redeem GRET it bought or received later. A wallet that only ever bought GRET on the market has to mint once first, which it cannot do while governance has minting paused (section 6.8) or while a frozen token is still active with a balance, since that mint must deposit it. This rule does not apply in a liquidation, where every holder can exit (chapter 8).

3.2 How much you receive

Call `redeem(shares, minAmountsOut, version)`. If there is a redemption fee, its part of your GRET goes to the fee recipient and only the rest is burned. For each active token you receive *burned GRET ÷ GRET in circulation × what the vault holds of that token*, rounded down. `minAmountsOut` sets the least you accept of each token, and `previewRedeem(shares)` shows the amounts in advance, after the fee. As in a mint, `version` ties your minimums to the basket you saw; if a token is excluded after your quote, a positive minimum for it makes the redemption revert.

> **Example**
>
> The vault holds 10 PAXG and 4 cbBTC, and 100 GRET are in circulation. You redeem 5 GRET with no fee: you receive 5% of each balance, 0.5 PAXG and 0.2 cbBTC, and the supply drops to 95 GRET.

3.3 The 1 GRET floor

A redemption cannot leave between 0 and 1 GRET in circulation: either everyone leaves, or at least 1 GRET remains. Without this floor, tiny leftovers could distort the basket and make minting unusable. A consequence: if someone keeps a tiny amount of GRET, the last large holder cannot redeem their final bit normally; it can be recovered through liquidation or by gathering the GRET in one wallet.

3.4 All or nothing, and the emergency exit

A normal redemption reads every active token's balance and sends every token in one operation. If a single token cannot be sent (its issuer froze the vault, for example), or its balance cannot even be read, the whole redemption reverts and you keep your GRET. The vault never pays from an old record of what it held.

That is why there is a second way out: `redeemSkipping(shares, minAmountsOut, skipped, version)`. It is the same redemption, but it leaves out the tokens you list. The vault does not even call them, not even to read a balance, so no token can block it, however it misbehaves. `previewRedeemSkipping(shares, skipped)` quotes it the same way. You give up your part of the skipped tokens: it stays in the vault for the other holders, or, if you were the last holder, for whoever makes the next first mint. Set the minimum of a skipped token to 0.

Example: leaving while a token is frozen

The vault holds 10 PAXG, 4 cbBTC and 100 SPYx, and 100 GRET are in circulation. SPYx's issuer freezes the vault, so a plain redemption reverts. You hold 10 GRET and call `redeemSkipping` listing SPYx:

- you receive 10% of the PAXG and cbBTC: 1 PAXG and 0.4 cbBTC;
- your 10 SPYx stay in the vault, so the remaining 90 GRET now share 100 SPYx, about 1.11 per GRET instead of 1.

If SPYx recovers, those who stayed keep your part. That is the price of leaving early.

Why the emergency exit matters

Your protection against a malicious rebalancer is the 7-day notice: you can leave before it takes over. Without `redeemSkipping`, a captured governance could make one token fail on purpose and trap everyone during those 7 days. With it, whoever minted can always leave with every token that still works. Holders who never minted are not covered: governance can pause minting so that they cannot qualify (section 6.8).

The pilot's local page offers only the normal mint and redemption. If it cannot read a balance, it stops with an error and never skips a token on its own; an exit with `redeemSkipping` has to be made separately, accepting the loss of the skipped share.

CHAPTER 4

Fees

Governance can charge a fee on mints and on redemptions. Both start at zero, are paid in GRET and can never exceed 5%.

4.1 The mint fee

You deposit the basket for all the GRET the mint issues, and the fee's share of that GRET goes to the fee recipient. With a 1% fee, minting 1 GRET gives you 0.99 GRET and the recipient 0.01. The minter pays the whole fee, so the value behind every other GRET does not change.

Income that has already arrived needs no fee to protect it: it is in the balances a minter pays for (section 2.1). Someone who mints just before a known future distribution and redeems after it still takes a share of it. A fee adds friction to that, but neither a fee nor a pause fully prevents it (section 5.3).

4.2 The redemption fee

The fee's share of the GRET you hand in goes to the recipient, and only the rest is burned and paid out in tokens. With a 2% fee, redeeming 1 GRET moves 0.02 GRET to the recipient and pays you the tokens for 0.98 GRET.

4.3 Limits and changes

- **Cap:** at most 5% each, written in the code.
- **Raising waits:** a raise is announced and can take effect only after 7 days, during the following 7 days; anyone can apply it then. A cut applies at once and cancels any pending raise.
- **Recipient:** chosen by governance, at once. Once set, it can never be empty again, so a fee always has somewhere to go.
- **Never charged** on liquidation exits, swaps or ordinary GRET transfers.

4.4 Side effects

- The recipient's GRET is ordinary GRET and stays in circulation. If the recipient holds less than 1 GRET, it counts against the 1 GRET floor, so the last holders cannot fully redeem until the recipient hands its GRET over. While the redemption fee is positive, redemptions never empty the supply, because each one leaves its fee in circulation. In the pilot the recipient would be the pilot wallet.
- The `Minted` and `Redeemed` events report net amounts; the fee appears in its own `FeeCharged` event.

CHAPTER 5

The basket and its backing

What backs GRET is simply what the vault holds of each active token, read from the token itself on every operation. There is no separate record to keep up to date.

5.1 Components

The basket's tokens are called components. There can be between 1 and 128. The pilot starts with 11: ITOTon, SPYx, EFAon, IEMGon, SGOVon, BILon, JAAAx, PAXG, cbBTC, VNQon and DBCon, listed with their addresses in section 13.4. The cap of 128 exists because mint and redemption visit every component in one transaction, and Ethereum limits the gas a single transaction can use (about 16.7M). In the tests, a mint and a redemption with 128 simple tokens each stay under 8M.

5.2 Active and excluded tokens

Every component is either active or excluded.

State	What it means	How a token gets there
Active	Its whole balance in the vault backs GRET. Mint asks for it, redemption pays it, swaps can buy or sell it, and the price guard counts it.	The tokens the vault is deployed with; any component governance activates with `activateComponent`, at once
Excluded	It does not count for mint, redemption or swaps, which do not even call it. Its balance stays in the vault, protected: governance cannot rescue it while GRET exists. In a liquidation it is claimed like any other.	A newly added token; an empty one governance deactivates (section 6.4); a written-off one (section 6.5)

A donation cannot activate an excluded token; only governance can. Activating a token makes its whole balance back the GRET in circulation at that moment, so whoever holds GRET then shares it, including anyone who minted while it was excluded (section 5.3).

5.3 Income and losses

Because the balance is the backing, anything that changes an active token's balance changes what stands behind every GRET, at once and in proportion, with nothing to call:

- **Income.** An issuer distribution paid as more of the same token, such as an xStocks multiplier increase, a positive rebase, or a donation, belongs to the holders as soon as it arrives. A minter who comes later pays for it.
- **Losses.** An issuer seizure, a negative rebase or an issuer fee lowers the balance, and every holder shares the loss in proportion from the next operation on.

What is still open is timing. Whoever holds GRET when value arrives shares it, so someone who mints just before a known future distribution, just before an excluded token is activated again, or just before another holder leaves a token behind with the emergency exit, shares in that value without having been there before. A pause stops only those who would mint after it. Governance keeps minting paused through the time a written-off token may recover, and considers a pause well before a known distribution (risk R-08 and commitments P-4 and P-5, in chapter 11).

Only the basket's own tokens count. A distribution paid in another token, an airdrop, or income the vault would have to claim from an issuer is not included automatically.

5.4 Tokens sent by mistake, and rescue

Governance can withdraw tokens with `rescue(token, to, amount)`, under strict limits:

- **A token outside the basket:** all of it, at any time. This includes dust left behind by a removed component.
- **A basket token, active or excluded:** only while no GRET exists and outside a liquidation, when there are no holders to protect.
- **A written-off token:** never, not even at zero supply.

So a basket token sent to the vault by mistake backs every GRET at once, and governance cannot return it while holders exist. If someone sends GRET itself to the vault's address, the transfer simply fails, so nothing is lost and governance never holds GRET.

CHAPTER 6

Governance

Governance is a single address that decides changes. In the pilot it is the pilot wallet; later it can be a multisig, a timelock or a vote. Whatever it is, the vault sets the limits.

6.1 What governance can and cannot do

It can

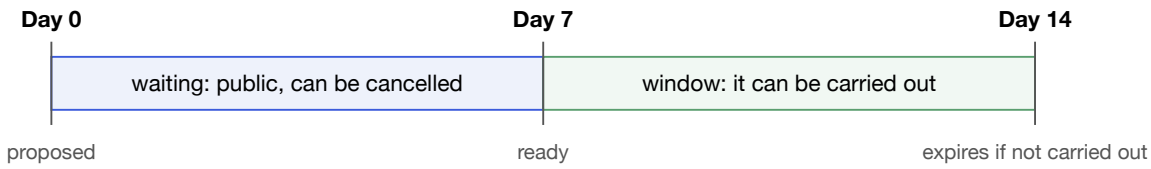
- Add, remove and write off components
- Activate and deactivate components
- Set the initial basket
- Set fees within 5%
- Appoint the rebalancer and the price guard; revoke the rebalancer
- Rescue tokens outside the basket
- Pause and resume minting
- Start a liquidation
- Appoint its successor

It cannot

- Withdraw basket tokens while GRET exists, or touch liquidation rights
- Take or remove a written-off token
- Mint GRET
- Change the contract's code
- Pause redemptions or reverse a liquidation
- Skip any of the waits
- Swap, unless it appoints itself rebalancer

The rules must live in the vault, not in the governance contract, because governance can be replaced. If the 7-day wait lived in governance, a new or captured governance could arrive without it. Think of the vault as a safe with a time lock: the board decides to open it, but the lock belongs to the safe, and it stays when the board changes.

6.2 How a change works: wait, then a window



Every important change follows the same calendar.

This applies to adding a token, writing one off, raising fees, appointing a rebalancer or a price guard, and handing over governance. Three of them (adding, writing off and applying a fee raise) can be carried out by *anyone* once ready: governance already decided when it proposed, so a voting

governance needs only one vote per change. The window keeps an old notice from being used by surprise months later.

Activating a token and deactivating an empty one take effect at once. Activation needs no wait of its own: the token already went through the notice of its addition or of its write-off. Deactivation only applies to a token the vault holds nothing of, or at most a few base units.

6.3 Adding a token

1. Governance announces it with `announceComponent(token)`. It must be a contract and not already in the basket.
2. From day 7 to day 14, anyone calls `addComponent(token)`. It joins at the end of the list, excluded, with zero initial amount. Even tokens sent to the vault before that do not count.
3. Governance activates it with `activateComponent(token)`, at once. While the vault holds none of it, mint does not ask for it; it starts counting once the rebalancer buys some with a swap (chapter 7).

Governance can cancel with `cancelComponentAnnouncement` during the wait. Tokens cannot be added during a liquidation or beyond 128. Once the token is active, governance should include it in the initial basket (commitment P-9).

6.4 Removing a token

1. The rebalancer sells the whole balance through swaps, until it is 0.
2. Governance calls `removeComponent(token)`. There is no wait, because a token the vault holds nothing of backs nothing. A rounding remainder is tolerated only for tokens with 18 decimals or more, and only up to 10 base units; any other token must be exactly empty.
3. The tokens after it move down one position. Any remainder becomes a token outside the basket, which governance may rescue.
4. Adding or removing a token raises `basketVersion` by one, so mints and redemptions prepared for the old list revert instead of applying their limits to the wrong token.

It cannot remove the last component, a written-off one, or anything during a liquidation. A mint or a donation that lands between the final sale and the removal leaves a balance that blocks it; another small swap sells it, which is why the final sale and the removal are best sent together (risk R-16). To keep a sold-out token in the basket without letting a stray unit make mints ask for it again, governance can instead call `deactivateEmptyComponent(token)`, under the same empty rule, and activate it again later.

6.5 Writing off a broken token

If an issuer freezes the vault, minting stops, every redemption has to skip that token with `redeemSkipping`, and the token cannot be sold because the vault cannot send it. The write-off lets the vault work normally again without it.

1. Governance proposes it with `proposeWriteOff(token)`. It judges that the token is broken; the vault does not check.
2. From day 7 to day 14, anyone calls `writeOffComponent(token)`. It makes no call to the token, so even a token that fails on every call can be written off.
3. The token is excluded and its initial amount drops to 0. Mint and redemption continue without it.

Why it cannot be used to steal

The tokens stay in the vault, the written-off token stays in the basket for good, and `rescue` and `removeComponent` refuse it forever. If it recovers, or if it was written off by mistake, `activateComponent` makes it back GRET again at once; in a liquidation, its recoveries can be claimed automatically, with no call at all. Holders lose that token's share only while it stays broken.

6.6 Appointing the rebalancer

There is no rebalancer at deployment. Governance proposes one with `proposeRebalancer(address)`, and that address accepts with `acceptRebalancer` between day 7 and day 14. Governance can revoke the rebalancer at once with `revokeRebalancer`, which also discards one still pending: taking power away never needs a wait.

Before any swap, governance must also appoint a price guard with `proposePriceGuard(address)`. After the same 7 days anyone applies it with `applyPriceGuard`, until day 14. Section 7.4 explains what it does.

6.7 Setting fees

Governance calls `proposeFees(mintFee, redeemFee)`, in basis points (100 = 1%). Lower values apply at once. Higher ones wait and are applied with `applyFees` between day 7 and day 14. Governance chooses the recipient with `setFeeRecipient`. Details in chapter 4.

6.8 Pausing the mint

Governance can pause minting, and resume it, at once with `setMintPaused`. Everyone who minted can still redeem, normally or with the emergency exit. A wallet that never minted, though, cannot qualify while the pause lasts: if it bought GRET, it has to sell it or wait. This is an accepted risk (R-07 in section 11.4): redemption stays for wallets that minted, and a pause that let new wallets mint would not serve its purpose.

The pause is useful during an incident, while something is investigated, throughout the time a written-off token may recover, and before a known distribution. It only stops those who would mint after it: anyone who minted before still shares in whatever arrives later (section 5.3).

There is no way to pause redemptions. That is deliberate: a pause on exits is exactly the power to trap holders during the 7-day notice of a new rebalancer. For real emergencies the tools are the write-off and the liquidation.

6.9 Handing over governance

The current governance calls `proposeGovernance(next)`, and the new one accepts with `acceptGovernance` between day 7 and day 14. A timelock with its own delay of up to 14 days can still accept in time if it schedules the acceptance right away.

6.10 From a wallet to a vote

For the vault, governance is just one address, and it can change: from the pilot wallet to a multisig, then a timelock, and finally a vote by VORA holders. Each step uses the handover of section 6.9, without touching the vault. These stages are a proposal, not a decision; chapter 10 explains each one and how voting would work.

Mind the names: the Governor's `propose` creates a vote; the vault's `announceComponent` starts a token's 7-day wait.

CHAPTER 7

The rebalancer and swaps

Rebalancing means changing what the vault holds. The rebalancer does it by exchanging one basket token for another, and mint and redemption follow on their own.

7.1 Why nobody sets "how much to deposit"

It might seem simpler for governance to declare "from today, 1 GRET costs 0.5 PAXG". It cannot work. Mint and redemption must follow what the vault holds, or new minters would take value from existing holders, and without prices the vault cannot tell whether two different baskets are worth the same.

The problem in numbers

The vault holds 10 PAXG and 10 GRET: 1 PAXG per GRET. If someone could mint 1 GRET paying 0.5 PAXG, there would be 10.5 PAXG for 11 GRET, about 0.95 PAXG each. The newcomer paid 0.5 and holds 0.95; each earlier holder lost 0.05.

7.2 What a swap does

The rebalancer calls `swap(tokenOut, amountOut, tokenIn, amountIn)`. Both amounts are fixed in the call: together they are the price.

1. The vault checks that GRET is in circulation, that both tokens are active, and that it holds at least `amountOut` of the outgoing token.
2. It collects `amountIn` of the incoming token from the rebalancer and measures what arrived, with the same tolerance as mint.
3. It asks the price guard. Only if the guard accepts does it send `amountOut` of the outgoing token.
4. Both balances change; the GRET supply does not.

Example

100 GRET are in circulation and the vault holds 10 PAXG and 0.5 cbBTC, so 1 GRET = 0.1 PAXG + 0.005 cbBTC. The rebalancer delivers 0.05 cbBTC and takes 1 PAXG. Now the vault holds 9 PAXG and 0.55 cbBTC, and 1 GRET = 0.09 PAXG + 0.0055 cbBTC. The next mint asks for exactly that.

At October 22 prices (PAXG ≈ USD 4,050, cbBTC ≈ USD 81,150), 1 PAXG ≈ 0.05 cbBTC, so the price was fair. Had the rebalancer delivered only 0.01 cbBTC, holders would have lost the difference.

7.3 Trust and its limits

The rebalancer chooses the price; the price guard bounds it. A malicious rebalancer could still sell a little cheaply or buy a worthless token that governance added and activated, but only within the guard's limits. The defenses are the price guard, the 7-day notice before any new rebalancer or guard acts, and the emergency exit of chapter 3.

- **Pilot:** the pilot wallet, with the Chainlink guard.
- **Later:** a bot or contract that finds the best price, still within the guard. It would be appointed through the same 7-day wait, without changing the vault.

7.4 The price guard

Every swap must be accepted by the price guard, a separate contract whose address the vault keeps. Governance appoints it like a rebalancer, with the 7-day wait, and without a guard there are no swaps. Mint, redemption and liquidation never ask it anything, so not even a broken guard can block an exit.

The first guard, `ChainlinkPriceGuard`, values both sides of each swap in dollars with Chainlink price feeds and refuses:

- a token with no feed, or whose price is stale, such as a stock feed outside market hours;
- a swap that loses more than 2% of what the vault sells;
- a swap that would make the week's swaps take more than 2% of the vault's value. Each loss counts as the share of the vault it takes at that moment, which is what every GRET loses, so money that comes in or leaves during the week does not stretch the limit. The vault is valued by what it actually holds of each active token just before the swap: the incoming payment, which has already arrived when the guard is asked, is taken out, and excluded tokens do not count. A token whose balance cannot be read counts as zero, which only makes the limit stricter.

> **Why a weekly limit**
>
> With only a per-swap limit, a stolen rebalancer key could trade the same value back and forth: USD 1,000 for 980, then 980 for 960, and so on. Fifty swaps, each within 2%, would leave 364. The weekly limit caps the total: in a week, all swaps together can take at most 2% of the vault, however many there are. An honest rebalance barely touches it: selling USD 300 at 0.5% below the price takes 0.15% of the vault.

The guard has no owner. Its feeds and limits are fixed when it is deployed, and changing any of them means deploying a new guard and waiting the 7 days. It is only as honest as its prices: a captured governance could appoint a guard that accepts everything, but the 7-day notice gives holders time to leave, as with a rebalancer. And it only sees what its feeds see: if a token trades far below the asset its feed prices, a swap that buys it looks fair and is not limited at all (R-03 in section 11.4).

The vault and its guard must be deployed as a pair: this guard is built for a vault that backs GRET with live balances.

Today Chainlink has feeds on Ethereum for five of the eleven tokens: SPYx (through SPY's price), SGOVon, BILon, PAXG and cbBTC. The other six cannot be swapped until feeds exist and a new guard is appointed. Stock feeds pause on weekends, so swaps of those tokens do too.

7.5 Funding a swap

Because the vault collects the incoming token first, the rebalancer must have it before the swap. There are three ways to do that without changing the vault:

- **Own funds, in steps.** Buy some cbBTC, swap it in, receive PAXG, sell the PAXG and repeat. You need the capital of one step, not of the whole rebalance. This is the pilot's way.
- **A flash loan.** A small rebalancer contract borrows the incoming token (for example, cbBTC from Morpho or Aave), swaps it into the vault, sells the token it receives on an exchange and repays the loan, all in one transaction. It needs no capital, only exchange liquidity for both tokens. The contract chooses the price, so it should deliver what the exchange pays, minus costs.
- **An auction.** Each bidder brings its own incoming token; the auction contract calls the swap and hands over what comes out. Bidders compete, which also gives better prices.

Swaps are closed during a liquidation and while no GRET exists. There is no cap on how much can be swapped, only on how much swaps can lose: a volume cap would slow down a stolen key, but also a legitimate large rebalance.

CHAPTER 8

Liquidation

Liquidation is the emergency exit for everyone. It ends normal operation for good and lets every holder take their share, token by token.

8.1 Starting it

Only governance can start a liquidation, with `activateLiquidation`, and only while GRET is in circulation. It is irreversible. It closes mint, normal redemption, swaps and basket changes, and it freezes the GRET supply at that moment as the denominator of every holder's right. It does not call any token, so a broken token cannot prevent it.

Every token in the basket counts, excluded and written-off ones included: a write-off never takes away liquidation rights. Each holder's right over a token is *GRET converted ÷ supply at activation* × *that token's basis*. The basis is what the vault holds of the token when it is first claimed successfully. It never shrinks afterwards. If more of the token arrives, so that what the vault holds plus what it has paid out exceeds the basis, the basis grows to match, and so does everyone's right, in proportion, including holders who already claimed. Converting early or late, all at once or in several steps, gives exactly the same share. What someone claims is recorded apart and never lowers the basis the others' shares are computed on.

8.2 Exiting token by token

1. Any holder, whether they minted or not, calls `convertToClaims(shares)`. Their GRET is burned and turned into rights over every token. This can be done in several steps, and unconverted GRET stays transferable.
2. For each token, the holder calls `claim(index, to, minAmount)`. Each token is claimed separately, and can be sent to another address if, for example, an issuer blocks the holder's own. `claimable(account, index)` shows what a claim would pay now.

No fee is charged on any of this.

8.3 When a token falls short or fails

- **A token fails to transfer, or its balance cannot be read:** that claim reverts and the right stays pending, ready to claim later. The other tokens are unaffected.
- **The vault holds less than everyone is owed:** each claimant is paid the same fraction, and their right is used up. `minAmount` lets a holder refuse a cut and wait, in case the balance recovers.
- **Tokens arrive later:** they first cover the rights still pending in full; anything beyond that raises everyone's rights in proportion, automatically. A cut already taken is not refunded on its own,

and governance cannot redirect anything, because no basket token can be rescued during a liquidation.

- **A loss before anyone claims a token:** its first claim simply sets a smaller basis, and every holder gets the same smaller share.

Example: a liquidation with a shortfall

Governance starts a liquidation with 100 GRET in circulation; the vault holds 10 PAXG and 100 SPYx. Alice holds 50 GRET, Bob 30 and Carol 20, and all three convert. Their rights: Alice 5 PAXG and 50 SPYx, Bob 3 PAXG and 30 SPYx, Carol 2 PAXG and 20 SPYx. Everyone claims their PAXG in full.

1. Alice claims her 50 SPYx first, which sets SPYx's basis at 100. 50 SPYx remain, owed to Bob and Carol.
2. SPYx's issuer seizes 25 SPYx: the vault holds 25 but owes 50. Carol claims now and is paid the same fraction anyone pending would get, 25 ÷ 50: she receives 10 SPYx, and her SPYx right is used up.
3. Bob sets `minAmount` to 30 and waits: the vault holds 15 and owes him 30, so a cut would pay him 15, and his claim reverts. His right stays pending.
4. The issuer returns the 25 SPYx. The vault holds 40, and what it holds plus what it has paid out is now 110, so the basis grows to 110 and every right grows by 10%: Bob's to 33, Alice's to 55 and Carol's to 22. Bob claims 33, Alice 5 more and Carol 2 more.

In the end Alice received 55 SPYx, Bob 33 and Carol 12: all 100 are paid out, but the recovery was shared by everyone instead of making Carol whole. The vault cannot correct this. The agreed policy is to settle it outside the vault, with separate funds or voluntary returns (commitment P-2).

CHAPTER 9

A year in the life of the vault

One story, from deployment to the end of the first year, with every number worked out. The basket is cut down to three tokens and the amounts are round, made-up figures, so the arithmetic stays easy to follow.

9.1 Deployment

The team deploys the vault with two components, PAXG and SPYx, and an initial basket of 0.01 PAXG and 0.1 SPYx per GRET. Governance is the team's wallet. There is no rebalancer, both fees are zero and minting is open. The vault is empty.

9.2 The first mint

The team mints 1 GRET, the only amount a first mint can ask for, and deposits exactly the initial basket. Its wallet can now redeem.

Supply	In the vault	Behind 1 GRET
1 GRET (team)	0.01 PAXG · 0.1 SPYx	0.01 PAXG · 0.1 SPYx

9.3 Alice joins

Alice mints 9 GRET and deposits nine times what stands behind 1 GRET: 0.09 PAXG and 0.9 SPYx. SPYx loses 2 base units on the way, so she receives 9 GRET minus a few wei; the story rounds that away.

Supply	In the vault	Behind 1 GRET
10 GRET	0.1 PAXG · 1 SPYx	0.01 PAXG · 0.1 SPYx

9.4 A new token joins

Governance wants Bitcoin in the basket.

- **Day 0.** It calls `announceComponent(cbBTC)`, proposes the team's wallet as rebalancer and proposes a Chainlink price guard. All three are public at once.
- **Day 7.** Someone calls `addComponent(cbBTC)` and `applyPriceGuard`, and the team's wallet accepts the rebalancer role. cbBTC joins the basket excluded, and governance activates it with `activateComponent(cbBTC)`. The vault holds no cbBTC yet, so nothing changes: mints still ask only for PAXG and SPYx.
- **Day 8.** The team buys 0.001 cbBTC on the market and swaps it in for 0.02 PAXG. At the day's prices both are worth about USD 81, so the price is fair, and the guard accepts it.

Supply	In the vault	Behind 1 GRET
10 GRET	0.08 PAXG · 1 SPYx · 0.001 cbBTC	0.008 PAXG · 0.1 SPYx · 0.0001 cbBTC

Nobody deposited or withdrew anything for the basket to change: the amounts behind each GRET simply followed what the vault holds.

9.5 A distribution arrives

In month four, SPYx pays a distribution by raising its multiplier: every SPYx balance grows by 1% (a round figure; a real quarterly dividend is smaller). The vault's 1 SPYx becomes 1.01 SPYx without any transaction.

Supply	In the vault	Behind 1 GRET
10 GRET	0.08 PAXG · 1.01 SPYx · 0.001 cbBTC	0.008 PAXG · 0.101 SPYx · 0.0001 cbBTC

Nobody calls anything. The team and Alice now have 0.101 SPYx behind each GRET, the next redemption pays it, and the next mint asks for it, so a newcomer pays for the distribution instead of sharing it.

9.6 A fee is turned on

Governance makes the team's wallet the fee recipient and proposes a 0.5% mint fee. Seven days later, Alice's bot applies it: anyone can. Then Bob mints 10 GRET. He deposits the basket for 10 GRET (0.08 PAXG, 1.01 SPYx and 0.001 cbBTC), receives 9.95 GRET, and the team's wallet receives 0.05 GRET.

Supply	In the vault	Behind 1 GRET
20 GRET (team 1.05, Alice 9, Bob 9.95)	0.16 PAXG · 2.02 SPYx · 0.002 cbBTC	0.008 PAXG · 0.101 SPYx · 0.0001 cbBTC

The value behind each GRET did not move: Bob paid for every share issued, including the fee's.

9.7 An issuer freezes the vault

In month seven, SPYx's issuer blocks the vault's address by mistake. The vault can no longer send or receive SPYx, so every mint reverts (it must pull SPYx) and every plain redemption reverts (it must send SPYx).

Governance pauses minting and proposes writing off SPYx. Alice does not want to wait: she calls `redeemSkipping` for her 9 GRET, listing SPYx. She receives 9/20 of the PAXG and cbBTC, 0.072 PAXG and 0.0009 cbBTC, and her 0.909 SPYx stay for the others.

Supply	In the vault	Behind 1 GRET
11 GRET (team 1.05, Bob 9.95)	0.088 PAXG · 2.02 SPYx (frozen) · 0.0011 cbBTC	0.008 PAXG · about 0.184 SPYx · 0.0001 cbBTC

Seven days after the proposal, Bob calls `writeOffComponent(SPYx)`. SPYx is excluded: the vault no longer counts it or calls it, while the 2.02 SPYx stay in the vault, out of governance's reach. Plain redemptions work again, without SPYx: 1 GRET now stands for 0.008 PAXG and 0.0001 cbBTC. Minting stays paused, because SPYx may come back.

9.8 The issuer lifts the freeze

Two months later, the issuer unblocks the vault. With minting still paused, governance calls `activateComponent(SPYx)`: the 2.02 SPYx that never left the vault count again at once. Then it resumes minting.

Supply	In the vault	Behind 1 GRET
11 GRET	0.088 PAXG · 2.02 SPYx · 0.0011 cbBTC	0.008 PAXG · about 0.184 SPYx · 0.0001 cbBTC

The team and Bob now have about 0.184 SPYx behind each GRET, up from 0.101: they kept Alice's share of SPYx, the price of her early exit. Had minting stayed open while SPYx was excluded, someone could have minted without depositing any SPYx and shared in it once it was activated again.

9.9 A notice that expires

In month eleven, governance proposes raising the mint fee to 1%, then changes its mind and forgets to cancel. The raise is ready on day 7 and nobody applies it. After day 14 it has expired: nobody can apply it later, and a new raise would need a new 7-day notice.

9.10 The year at a glance

When	What happened	Supply
Day 0	Deployment; the team mints the first 1 GRET	1
Month 2	Alice mints 9 GRET	10
Month 3	cbBTC announced, added and activated; the team's wallet becomes rebalancer and swaps PAXG for cbBTC	10
Month 4	A 1% SPYx distribution counts at once, with nothing to call	10
Month 5	0.5% mint fee; Bob mints 10 GRET and the fee goes to the team's wallet	20
Month 7	SPYx frozen; minting paused; Alice leaves skipping SPYx; SPYx written off	11
Month 9	Freeze lifted; SPYx activated again; minting resumed	11
Month 11	A fee raise is proposed, never applied, and expires	11

CHAPTER 10

Governance in practice: from a wallet to a VORA vote

The vault only checks which address is governance. Who stands behind that address, and how they decide, can grow over time without changing the vault. This chapter walks through each proposed stage, ending with on-chain voting by VORA holders using OpenZeppelin's standard contracts. None of these stages is decided or implemented yet.

10.1 Four stages

Stage	How a decision is made	What it takes
1. Pilot wallet	One signature, at once. The pilot's stage.	Nothing
2. Safe	Several signers must approve, for example 2 of 3, at once.	A Safe created at app.safe.global; no code
3. Safe + timelock	The Safe schedules each action, and it runs only after a public delay.	OpenZeppelin's `TimelockController`, already tested with the vault
4. VORA vote + timelock	VORA holders vote; what passes waits in the same timelock, then runs.	OpenZeppelin's `Governor`, and VORA built on `ERC20Votes`

Each move from one stage to the next is a governance handover (section 6.9): the current governance proposes the next address, and that address accepts between day 7 and day 14.

10.2 Stage 2: a multisig

A Safe is a wallet that needs several signatures to act: with 2 of 3, any two of three signers must approve every transaction. Losing one key no longer locks governance, and stealing one key is not enough to control it. A Safe can also bundle several calls into one transaction, which all succeed or all fail together.

To move there, the pilot wallet calls `proposeGovernance(safe)`, and the Safe calls `acceptGovernance` between day 7 and day 14. Decisions are still immediate: the Safe acts as soon as enough signers approve.

10.3 Stage 3: a timelock

A `TimelockController` is a contract that holds a queue of actions. Every action is *scheduled* first and can be *executed* only after a minimum delay, so anyone watching the chain sees it coming. The timelock becomes the vault's governance; the Safe keeps the power to schedule.

Timelock role	What it allows	Proposed holder
Proposer	Schedule actions (each proposer is also made a canceller)	The Safe
Executor	Run an action once its delay has passed; can be open to anyone	The Safe
Canceller	Remove a scheduled action before it runs	The Safe
Admin	Change the roles	Nobody: the timelock manages itself

The delays add up. With a 2-day timelock, adding a token takes 2 days for the timelock plus the vault's own 7 days: holders get 9 days of notice. Liquidation, which the vault does not delay, would then wait the timelock's 2 days.

One detail of the handover

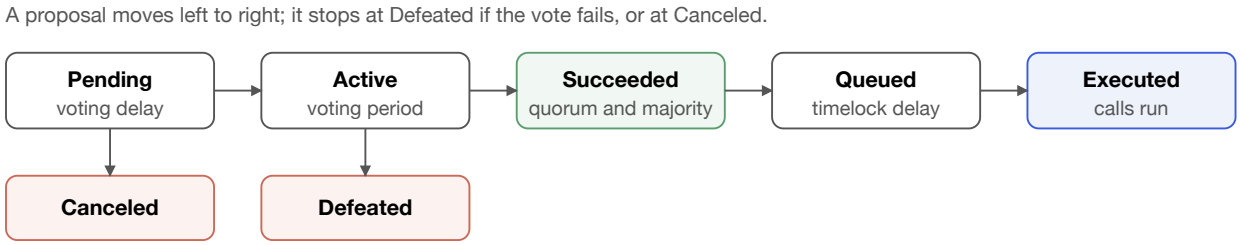
The timelock must accept governance itself, which means the call `acceptGovernance` has to go through its own queue and land between day 7 and day 14. The Safe should schedule it right after proposing the handover, with a timelock delay of at most 14 days.

10.4 Stage 4: the voting token

VORA is the project's proposed voting token; GRET itself never votes. For the Governor to count votes, VORA must be built on OpenZeppelin's `ERC20Votes`, which keeps a history of each holder's voting power.

- **Voting power must be activated.** Holding VORA is not enough: a holder must *delegate* its votes, to itself or to someone it trusts. Until then, its tokens do not count.
- **Votes are counted at a fixed moment.** Each proposal takes a snapshot of voting power when voting opens. Buying, borrowing or receiving VORA afterwards changes nothing, so a flash loan cannot swing a vote.
- **Delegates.** Holders who do not follow every proposal can delegate to a representative and take the votes back at any time.

10.5 A proposal from start to finish



The states of an OpenZeppelin Governor proposal, when it runs through a timelock.

1. **Propose.** A holder with enough voting power (the *proposal threshold*) submits a list of exact calls, for example "call `vault.announceComponent(X)`", with a description. The calls cannot be changed afterwards.
2. **Wait (Pending).** The *voting delay* gives holders time to read the proposal and delegate. Voting power is fixed when it ends. The proposer can still cancel during this phase.
3. **Vote (Active).** During the *voting period*, each holder votes For, Against or Abstain with the power it had at the snapshot.
4. **Count.** The proposal succeeds if For plus Abstain reach the *quorum* (a share of all VORA at the snapshot) and For is strictly larger than Against. Otherwise it is defeated.
5. **Queue.** A successful proposal is sent to the timelock, where its calls wait the timelock's delay in public.
6. **Execute.** After the delay, the calls run from the timelock, in order and all together, exactly as voted.

10.6 Settings to choose

These numbers are set when the Governor is deployed and can later be changed only by a vote. The values below are common starting points, not decisions.

Setting	Example	What it protects against
Voting delay	1 day	Surprise votes: time to read a proposal and delegate before power is fixed
Voting period	5 to 7 days	Votes that pass before most holders notice
Proposal threshold	0.5% to 1% of VORA	Spam proposals
Quorum	4% to 10% of VORA	A small group deciding while everyone else is absent
Timelock delay	2 to 7 days (at most 14)	Changes that run before holders can react

Late-quorum extension	2 days	A large holder tipping the vote at the last minute; the period is extended so others can answer
Proposal guardian	The Safe	A malicious proposal that passes: the guardian can cancel it at any stage, but can never pass anything

The last two are optional OpenZeppelin extensions (`GovernorPreventLateQuorum` and `GovernorProposalGuardian`).

10.7 Who holds what at stage 4

- The vault's governance is the timelock.
- The Governor is the timelock's proposer and canceller. The executor can be the Governor or open to anyone.
- The Safe stops being a proposer. OpenZeppelin warns that any extra proposer could act as the timelock without a vote. The Safe can stay as proposal guardian, with power only to cancel.
- The timelock has no admin: only a vote can change its roles.

10.8 Example: adding a token by vote

Day	What happens
0	A VORA holder proposes one call: `vault.announceComponent(X)` . Everyone can read it on the chain and on Tally.
1	The voting delay ends and voting power is fixed. Voting opens.
6	Voting closes: quorum reached and more For than Against. Anyone queues it in the timelock.
8	The timelock's 2 days pass. Anyone executes it: the vault's 7-day announcement starts.
15 to 22	Anyone calls `addComponent(X)` , with no further vote. X joins the basket excluded.
After that	A second proposal, voted the same way, carries `activateComponent(X)` and `setInitialAmounts` to include X in the initial basket. Once it runs, X counts and the rebalancer can buy it.

A proposal can carry several calls that run together, in order, in one transaction, as the second one above does. Votes are public for days, which is fine for most changes. Activating a token that was written off is the exception: anyone who mints while that vote is public shares in the recovery, so minting should already have been paused for the whole exclusion, not just in the same proposal (commitment P-5).

10.9 What can go wrong

Risk	Defense

Someone buys or borrows VORA to pass a harmful proposal	The snapshot ignores power acquired after voting opens; the quorum, the voting period, the timelock delay and the guardian give time to react; the vault's own waits and the emergency exit still apply
Few holders vote	The quorum; delegation to active representatives
A harmful call is hidden in a proposal	Proposals list exact calls; interfaces such as Tally decode them; the timelock delay leaves time to inspect and cancel
A last-minute swing	The late-quorum extension
The guardian misbehaves	It can only cancel, never pass; a vote can replace it

Whatever happens in the governance layer, the vault's limits stay in place: waits, caps, the fee ceiling, the ban on taking basket tokens while GRET exists or written-off tokens ever, and the exits of chapters 3 and 8.

10.10 Tools

- **Tally** works with OpenZeppelin's Governor: it lists proposals, decodes their calls, and lets holders delegate, vote, queue and execute from the browser.
- **Safe** provides the multisig interface for stages 2 and 3, including batches of calls.
- **Snapshot** offers free off-chain polls. They are useful to sound out opinion but do not bind the vault unless a separate tool connects them to a Safe; they are not needed here.

CHAPTER 11

Safety

No contract is risk-free. This chapter separates what the code enforces from what depends on people and issuers, and lists every risk that was accepted and why.

11.1 What the code guarantees

- Nobody issues GRET without depositing the basket: governance has no mint.
- While GRET exists, basket tokens leave only when GRET is burned or another active token is delivered in exchange. Governance can rescue a basket token only when no GRET exists, and never during a liquidation.
- Income that arrives as balance of an active token counts for every holder at once, and a later minter pays for it; nobody has to record it.
- A redemption or a claim never pays from an old record of what the vault held: if a token's balance cannot be read, the operation reverts and nothing is consumed.
- The code never changes: no upgrades.
- A new rebalancer, price guard, token, governance, write-off or fee raise waits 7 days, visible on chain, and expires 7 days after that.
- Fees never exceed 5% each.
- Governance can never take or remove a written-off token.
- Redemption can never be paused, and whoever minted can always leave with every token that still works.
- The limits you sign hold: if a token is added or removed in between, your transaction reverts; if one is activated or excluded, your input and output limits still bound what you pay and accept.
- Every swap passes the price guard; with the Chainlink guard, swaps lose at most 2% each and 2% of the vault per week.
- In a liquidation, every holder can exit token by token, even if some token fails, and later income is shared automatically.

11.2 What it does not

- **The price of swaps, fully.**

  A malicious rebalancer can still take up to the guard's weekly limit until it is revoked, or more if a token trades below the asset its feed prices, and a captured governance can appoint a guard that accepts everything; the 7-day notice is the defense against that. Six tokens have no price feed and cannot be swapped.
- **An exit for every holder during the notice.**

  Only wallets that minted can redeem. The others have to mint first, which governance can block by pausing minting; otherwise they depend on selling their GRET or on a liquidation, which only governance can start.
- **The issuers.**

  All 11 tokens are upgradeable; their issuers can freeze, charge or change them.
- **The timing of future value.**

  Whoever mints before a future distribution, or before an excluded token is activated again, shares in it. Keeping minting paused through a recovery covers the second case; nothing in the vault fully covers the first.
- **Mistakes and targeted compensation.**

  A basket token sent to the vault by mistake cannot be returned while GRET exists. In a liquidation, a recovery is shared by every holder, not returned to whoever took a cut.
- **The absence of bugs.**

  The contract has had internal reviews, not an independent audit.
- **Everything else that was weighed and accepted:**

  section 11.4.

11.3 The pilot

In the pilot, governance is the pilot wallet. Whoever obtains its key can start a liquidation, rescue tokens outside the basket, activate tokens and, after a 7-day wait visible on chain, appoint themselves rebalancer and exchange the basket for anything. If the key is lost, nobody can liquidate, activate a token or change the basket, although income keeps counting and holders can still mint and redeem. Before third-party funds, governance should move to a Safe and the contract should be audited.

11.4 Accepted risks, one by one

These are the known risks that were weighed and accepted, each with its reason. Problems that were found and fixed are not listed, except two that keep their IDs as closed entries. Each risk keeps a stable ID, used in `docs/ACCEPTED_RISKS.md`, which also names the test that reproduces it. The P numbers point to the commitments in section 11.5.

ID	What can happen	Worst case	Why it is accepted, and what limits it
Governance and rebalancing			
R-01	Governance holds broad powers, and its key can be stolen or lost.	After a 7-day notice, the basket swapped for worthless tokens. If the key is lost, nobody can liquidate, activate tokens or change the basket.	Someone has to manage the basket and emergencies, and every power that can take value waits 7 days in public. Wallets that minted can leave during the notice. A Safe before third-party funds (P-12).
R-02	The rebalancer chooses the price of each swap, within the guard's limits.	2% of the value behind each GRET per week, or 4% across a change of week, until the rebalancer is revoked; deposits and withdrawals do not stretch it.	Exact prices are impossible with thin liquidity, and a volume cap would also block an honest rebalance. Revoking takes effect at once (P-6, P-8).
R-03	The guard is only as good as its feeds. A feed of the underlying keeps pricing a token that trades far below it, so buying that token at the feed price records no loss. Feeds also move in steps of 0.5% to 2%, six tokens have no feed, and stock feeds stop on weekends.	The healthy tokens the guard lets be sold, swapped for the cheap token until the rebalancer is revoked; not bounded by R-02. Half of the basket cannot be rebalanced.	The available feeds price the underlying or the fund; a market price of each token would need another oracle. Watch market prices against the feeds and revoke at once (P-8); a new guard when feeds appear (P-10).
R-04	Every balance of a basket token backs GRET, including mistaken transfers, and governance cannot rescue it while GRET or liquidation rights exist. In a liquidation, a balance that comes back after claims took a cut is shared by every holder.	Holders who took a cut are not made whole; a mistaken deposit of a basket token cannot be returned.	Per-holder compensation would need more liquidation accounting, and automatic allocation takes discretion over basket income away from governance. Settle outside the vault (P-1, P-2, P-3, P-7).

ID	What can happen	Worst case	Why it is accepted, and what limits it
R-05	An issuer gives a token a second address over the same balance, and governance rescues it through that address.	A whole component.	It needs both an issuer change and a dishonest governance; none of the 11 tokens has a second address (P-10).
R-06	Governance writes off a healthy token.	Whoever mints before it is activated again shares in it (R-08).	Governance gains nothing: it can never rescue or remove the token, and `activateComponent` undoes the exclusion (P-5).
Entering and exiting			
R-07	While minting is paused, or a token is frozen, a holder who never minted cannot qualify for redemption.	During a hostile notice, such holders can only sell, hand their GRET to a wallet that minted, or wait for a liquidation.	Redemption stays for wallets that proved access to the whole basket, and a pause that let new wallets mint would not serve its purpose. In the pilot, the only holder minted.
R-08	Whoever mints before a future distribution, a favorable swap, the activation of an excluded token, or another holder's skipping exit shares in that value. Income already in the vault cannot be captured this way: a minter pays for it.	A large share of a future distribution, a recovery or abandoned tokens.	Giving it only to earlier holders needs record dates and more accounting. Pauses well before known events and through recoveries; a fee adds friction only (P-4, P-5, P-6).
R-09	The 1 GRET floor holds back the last holders when someone keeps a fraction of GRET (a tiny mint, or fees held by the recipient), or while a redemption fee is positive.	Just under 1 GRET held back, or 1 GRET while the redemption fee is positive; a liquidation releases it.	The floor stops rounding from distorting the basket; a seed locked forever would cost about the same. In the pilot the recipient is the pilot wallet.
R-10	Minimums bound what the vault sends, not what arrives.	Less than the minimum received, or nothing from a token that fakes its transfers.	Checking the recipient would add calls to every exit, and a token that lies about transfers can lie about balances too (P-10).
R-11	A token may arrive up to 10 base units short; after a severe loss, that can cost a minter more of the other tokens.	A minter overpays; for cbBTC, 10 units are about one cent.	The xStocks lose 2 to 3 units per transfer. `minShares` lets the minter refuse.
Issuers and tokens			
R-12	Issuers can pause, freeze, seize or change their tokens.	A token lost to holders.	It comes with tokenized assets. `redeemSkipping`, a write-off and liquidation limit the damage (P-10).

ID	What can happen	Worst case	Why it is accepted, and what limits it
R-13	A token upgraded to take more from the vault than it sends.	The remaining holders pay the difference, and the last one cannot withdraw that token.	None does it today, and a check after every transfer was rejected. Holders who minted skip it; governance writes it off or sells it.
R-14	**Closed.** A token's `balanceOf` breaks while its transfers work. The vault never pays from an old record, so the first to leave can no longer take full payment and leave the shortfall to the last.	That token cannot leave normally until it recovers.	Holders who minted skip it with `redeemSkipping`; a write-off excludes it; in a liquidation the other tokens stay claimable.
R-15	A token becomes expensive to call, or calls back into the vault.	Normal redemption no longer fits in one transaction, or swaps stop, since the guard reads every priced balance.	No cap can bound outside code. `redeemSkipping` never calls the skipped tokens, every operation on the basket is guarded against reentrancy, and the accounting views refuse to answer in the middle of one.
The basket			
R-16	A remainder after the sale of a whole balance, or a donation of even one base unit, makes an active token required again for minting, and blocks its removal beyond the small tolerance of section 6.4.	Another sale; for a token that cannot be sold, a write-off that keeps its slot for good.	A swap must not change its price after it was set, and every active balance backs GRET, unsolicited ones included. Empty tokens can be deactivated (P-11).
R-17	Written-off tokens keep their place forever.	After more than a hundred write-offs, no room for new tokens.	Keeping them protects their balance and lets a recovery reach the holders, also in a liquidation.
R-18	Any contract can be announced as a new token, and activation does not check it.	A useless or hostile token, after 7 days of notice.	The vault cannot judge tokens; the notice gives time to leave (P-10).
R-19	The initial amounts go stale after basket changes.	If everyone leaves, nobody can mint again until governance updates them.	They matter only at zero supply (P-9).
R-20	A tiny balance of a rounding token might make small mints revert; this was never reproduced.	Small mints revert until the remainder is sold.	Whole balances are sold and the token removed or deactivated (P-11).

ID	What can happen	Worst case	Why it is accepted, and what limits it
Liquidation			
R-21	When every pending payout of a token rounds to zero, nobody can claim what is left.	A few base units: a fraction of a cent with the current tokens.	Pooling rights would add code for dust.
R-22	Rights over a token that never moves again, and GRET held by contracts that cannot convert it, are never paid.	Those rights stay pending for good.	A liquidation cannot force a token to move, and contracts must provide their own exit.
R-25	**Closed.** A liquidation started with no GRET in circulation would leave later recoveries with nobody to claim them.	None in this version.	`activateLiquidation` refuses it, so an empty vault stays in normal mode.
Limits of the design			
R-23	The vault cannot change: no upgrades, no new rules such as an allowlist, and only what arrives as balance of an active token counts automatically.	A bug or a new rule needs a new vault, which holders join voluntarily.	Any power to change the vault would also be a power to act against holders. Tests, reviews, an audit before third-party funds, and a pilot with own funds.
R-24	Fees round down.	Operations of a few wei pay no fee.	Avoiding it costs more gas than it saves.

11.5 What governance commits to

Many of the limits above depend on how governance behaves. The vault does not enforce these commitments; they are the agreed policy. `docs/ACCEPTED_RISKS.md` also has P-13, which applies only to the vault deployed on October 27, 2026.

ID	Commitment	For
P-1	Watch the vault's actual balances and the issuers' incidents. No sync is needed, but losses and recoveries must be reconciled for P-2 and P-7.	R-04
P-2	In a liquidation, work out from the `Claimed` events who took a cut and who was paid more after a recovery, and settle it outside the vault, with separate funds or voluntary returns. Record every settlement.	R-04
P-3	Watch issuer distributions and compare balances against mints, redemptions, swaps and claims. Distributions paid in other tokens, or that must be claimed from the issuer, are not automatic.	R-04

ID	Commitment	For
P-4	Watch announced distribution schedules and, before a known distribution, consider pausing minting early enough to matter. Approve exact amounts and refresh quotes after approvals.	R-08
P-5	Write off only a token that stopped working, and cancel if it recovers. Pause minting before the write-off can run, keep it paused while the token may recover, and activate it again explicitly. Before the last holder leaves skipping a token, pause and decide who owns what stays behind.	R-06, R-08
P-6	Send every swap through a private RPC.	R-02, R-08
P-7	If a removed token gets its balance back before a liquidation, add it again and activate it. During a liquidation, rescue it and pay the entitled holders outside the vault.	R-04
P-8	Watch the guard's `LossRecorded` events, and the market price of every priced token against its feed. Revoke the rebalancer after any loss nobody expected, or when a token trades away from its feed. Deploy the vault with a guard built for live balances.	R-02, R-03
P-9	Update the initial amounts with every basket change, activation and exclusion included. Before a restart from zero supply, check every balance already in the vault.	R-08, R-19
P-10	Admit tokens carefully and watch their issuers: no transfer fees, callbacks or second addresses; 8 decimals or more; a Chainlink feed if the token must be rebalanced.	R-03, R-05, R-10 to R-15, R-18, R-21
P-11	Sell remainders, and remove or deactivate sold-out tokens promptly, in the same transaction as the final sale when possible.	R-16, R-20
P-12	Before third-party funds, move governance to a Safe with a tested recovery procedure.	R-01

CHAPTER 12

Questions and answers

Short answers to the questions people ask first, with a pointer to where the full explanation is.

12.1 Can governance take my tokens?

Not directly. It cannot withdraw basket tokens while GRET exists, mint GRET or touch liquidation rights. Its most dangerous power is indirect: appointing a rebalancer, and a price guard that accepts any price. That takes a 7-day public notice, during which anyone who minted can redeem, skipping any token that was made to fail (chapters 3 and 7). With the Chainlink guard in place, even a stolen rebalancer key can lose at most 2% of the vault per week.

12.2 I bought GRET on an exchange. Why can't I redeem it?

Normal redemption is only for wallets that have minted at least once. Mint any amount once from the same wallet and you can redeem everything you hold, including what you bought. While governance has minting paused you cannot do that first mint, so you have to wait or sell. In a liquidation, every holder can exit without this condition (sections 3.1 and 8.2).

12.3 What happens if an issuer freezes the vault?

Minting stops and plain redemptions revert, because that token cannot move. Holders who minted can leave at once with `redeemSkipping`, giving up their part of the frozen token; those who never minted cannot qualify until the token is written off, because their first mint would have to deposit it. Governance can write it off after 7 days so the vault works normally again, and activate it again at once if the freeze ends (sections 3.4 and 6.5).

12.4 What if an issuer takes tokens from the vault or charges a fee?

The vault always uses what it actually holds, so the loss is shared by every holder in proportion from the next operation on; nobody has to record it. A token that charges a fee on transfers makes mints revert for it, so it should not be in the basket (section 5.3).

12.5 Can the basket change without me knowing?

A new token always waits 7 days in public before it can join, and then counts only once governance activates it; a new rebalancer waits 7 days too. Once a rebalancer is in place, though, its swaps take effect at once: the proportions can change from one block to the next. What never changes without notice is who is allowed to swap (sections 6.3 and 7.2).

12.6 Why is there a 1 GRET floor?

Without it, someone holding almost everything could redeem down to a few wei, and the rounding leftovers would distort the basket so much that minting became unusable. The cost is that the last bit of the last large holder may need a liquidation, or the GRET gathered in one wallet, to come out (section 3.3).

12.7 Does GRET have a price?

Not inside the contract, which only counts tokens. GRET is worth whatever stands behind it, valued at the market prices of the basket's tokens. If it trades on an exchange, its market price can drift from that value, which is what makes minting and redeeming worthwhile for arbitrageurs.

12.8 Can the contract be upgraded?

No. Its code is fixed for good. Everything that can change (the basket, fees, the rebalancer, governance) changes within rules the code already contains. Anything beyond that requires a new vault, and holders would move voluntarily by redeeming from the old one and minting in the new one.

12.9 Do I earn anything by holding GRET?

GRET has no yield of its own. Issuer distributions that arrive as balance of an active token, such as the xStocks multiplier increases, count for all holders as soon as they arrive, with nothing to do. Distributions paid in other tokens, or that must be claimed from the issuer, are not included automatically. Value otherwise moves with the prices of the basket's tokens.

12.10 Can I lose money?

Yes. The basket's tokens can lose value; an issuer can freeze, seize or change its token; a rebalancer can trade at a bad price; and the contract can have bugs, since it has had internal reviews but no independent audit yet (chapter 11).

12.11 What if the pilot wallet's key is lost or stolen?

Lost: nobody can change the basket, activate a token, rescue or start a liquidation, but holders can still mint and redeem, and income keeps counting. Stolen: the thief can start a liquidation, rescue tokens outside the basket and, after a public 7-day notice, appoint a rebalancer and swap the basket away. Holders who minted can leave during that notice (section 11.3).

12.12 I sent a basket token to the vault by mistake. Can I get it back?

Not while GRET exists. Any balance of a basket token backs every GRET at once, and governance cannot rescue basket tokens while there are holders. A token outside the basket, by contrast, can be rescued and returned (section 5.4).

CHAPTER 13

Reference

Every function and every fixed number in one place.

13.1 Functions

Function	Who	What it does	Condition
Holders			
`mint`	Anyone	Deposits the basket and receives GRET, minus the mint fee	No liquidation; not paused; current basket version
`redeem`	Wallets that minted	Pays the fee in GRET, burns the rest, receives every active token	Supply left 0 or ≥ 1 GRET; current basket version
`redeemSkipping`	Wallets that minted	Same, without the listed tokens, which are not called	Their minimum must be 0; current basket version
Basket			
`announceComponent`	Governance	Announces a new token	—
`cancelComponentAnnouncement`	Governance	Discards the announcement	—
`addComponent`	Anyone	Adds the token, excluded	Day 7 to 14; max 128
`activateComponent`	Governance	Makes a token's whole balance back GRET	Immediate; no liquidation
`deactivateEmptyComponent`	Governance	Excludes an empty token	Balance 0, or ≤ 10 base units with ≥ 18 decimals
`removeComponent`	Governance	Removes a token	Same empty rule; not the last; never written off
`proposeWriteOff`	Governance	Proposes writing off a broken token	Governance judges it
`cancelWriteOff`	Governance	Discards the proposal	—
`writeOffComponent`	Anyone	Excludes the token and marks it written off for good	Day 7 to 14; token active
`setMintPaused`	Governance	Pauses or resumes minting	Immediate

`setInitialAmounts`	Governance	Sets the first mint's basket	Used only at zero supply; current basket version
`rescue`	Governance	Withdraws tokens outside the basket, or basket tokens while no GRET exists	Never written off; no basket token during a liquidation
Rebalancer			
`swap`	Rebalancer	Delivers one active token, takes another	No liquidation; GRET in circulation; a price guard
`proposeRebalancer`	Governance	Proposes a rebalancer	—
`acceptRebalancer`	Proposed address	Takes over the role	Day 7 to 14
`cancelRebalancerProposal`	Governance	Discards the proposal	—
`revokeRebalancer`	Governance	Removes the rebalancer and any pending one	Immediate
`proposePriceGuard`	Governance	Proposes the contract that must accept every swap	—
`applyPriceGuard`	Anyone	Applies the proposed guard	Day 7 to 14
`cancelPriceGuardProposal`	Governance	Discards the proposal	—
Fees			
`proposeFees`	Governance	Sets mint and redemption fees	Max 5%; raises wait
`applyFees`	Anyone	Applies a proposed raise	Day 7 to 14; needs a recipient
`cancelFeesProposal`	Governance	Discards a proposed raise	—
`setFeeRecipient`	Governance	Chooses who receives the fees	Immediate
Governance and liquidation			
`proposeGovernance`	Governance	Proposes the next governance	—
`acceptGovernance`	Proposed address	Takes over the role	Day 7 to 14
`cancelGovernanceProposal`	Governance	Discards the proposal	—
`activateLiquidation`	Governance	Starts the irreversible liquidation	GRET in circulation

`convertToClaims`	Any holder	Burns GRET in exchange for rights	During liquidation
`claim`	Whoever converted	Withdraws one token	During liquidation
Reading, free of charge			
`previewMint` , `previewRedeem`	Anyone	Exact amounts a mint asks for or a redemption pays, after fees	No liquidation
`previewRedeemSkipping`	Anyone	The same for the emergency exit, without calling the skipped tokens	No liquidation
`reserves`	Anyone	The live balance of every active token; in a liquidation, each token's basis	Fails if a balance cannot be read
`claimable` , `pendingRights`	Anyone	What a claim of one token would pay now, and the right behind it	Reads only that token

13.2 Fixed numbers

Constant	Value	Meaning
`INITIAL_SHARES`	1 GRET	What the first mint issues
`MIN_SUPPLY`	1 GRET	The floor a redemption cannot cross, except to zero
`MAX_COMPONENTS`	128	Most tokens in the basket
`MAX_RECEIPT_SHORTFALL`	10 base units	How short a token may arrive on mint or swap
`MAX_REMOVAL_DUST`	10 base units	Most a token with 18 decimals or more may still hold when it is removed or deactivated; other tokens must be empty
`BASKET_CHANGE_DELAY`	7 days	Wait for a new token, rebalancer, price guard or write-off
`GOVERNANCE_TRANSFER_DELAY`	7 days	Wait for a new governance
`FEE_INCREASE_DELAY`	7 days	Wait for a fee raise
`EXECUTION_WINDOW`	7 days	Time to carry out a ready proposal before it expires
`MAX_SWAP_LOSS` (guard)	200 (2%)	Most a single swap can lose, against the feeds
`MAX_WEEKLY_LOSS` (guard)	200 (2%)	Most all swaps in a week can lose, of the vault's value
`MAX_FEE`	500 (5%)	Highest mint or redemption fee, in basis points

13.3 Glossary

Term	Meaning
Active component	A basket token whose whole balance in the vault backs GRET: mint asks for it, redemption pays it and swaps can trade it.
Backing	What stands behind GRET: the vault's live balance of every active component.
Base unit	The smallest amount of a token. Most tokens have 18 decimals, so 1 token is 10^{18} base units; cbBTC has 8. For GRET the base unit is also called a wei.
Basis point	One hundredth of a percent: 100 basis points = 1%, 500 = 5%.
Basket	The set of tokens the vault holds; each is a *component*.
Basket version	A counter that grows whenever a token is added or removed. Mints and redemptions state the version they were prepared for, so a changed list makes them revert. Activating or excluding a token does not change it.
Burn	Destroying GRET, which lowers the supply. Redemptions and conversions to claims burn GRET.
Claim	In a liquidation, withdrawing one token against the rights obtained by converting GRET.
Delegation	Assigning one's VORA voting power to oneself or to a representative; tokens that are not delegated do not vote.
Excluded component	A basket token that does not count for mint, redemption or swaps: just added, deactivated while empty, or written off. Its balance stays protected and is claimed in a liquidation.
Fee recipient	The address that receives the mint and redemption fees, in GRET.
Flash loan	A loan taken and repaid within one transaction; if it is not repaid, the whole transaction is undone.
Governance	The address that decides changes within the vault's rules: in the pilot, the pilot wallet.
Governor	OpenZeppelin's standard voting contract, proposed for VORA holders.
Initial basket	The amounts the first mint deposits for exactly 1 GRET, used whenever the supply is zero.
Issuer	The company behind a component token (Ondo, Backed, Paxos, Coinbase), able to pause, freeze or upgrade it.
Liquidation	The irreversible emergency mode in which holders exit token by token.
Liquidation basis	The amount of a token that liquidation rights are computed on: what the vault holds at the token's first successful claim, raised later whenever what it holds plus what it has paid out exceeds it.

Mint	Creating GRET by depositing the basket.
Multisig (Safe)	A wallet that needs several signatures, for example 2 of 3, to act.
Price feed	A Chainlink contract that publishes an asset's price on the chain and updates it on a schedule or when the price moves.
Price guard	The contract that prices every swap and refuses the ones that lose too much. It has no owner; changing it takes the 7-day wait.
Proposal threshold	The voting power needed to submit a proposal.
Quorum	The share of all voting power that must vote For or Abstain for a proposal to be valid.
Rebalancer	The one address allowed to swap one component for another.
Redeem	Handing in GRET for the proportional part of every active component.
Rescue	Governance withdrawing a token outside the basket, or a basket token while no GRET exists; never a written-off token.
Reserve	Another word for the backing of one token: the vault's live balance of an active component.
Snapshot (voting)	The moment at which a proposal fixes each holder's voting power.
Supply	The total GRET in circulation.
Swap	The rebalancer delivering one active component and taking another, at amounts it sets.
Timelock	A contract that makes every action wait a public delay before it runs.
VORA	The project's proposed voting token for governance. GRET itself does not vote.
Write-off	Excluding a broken component after a 7-day notice, so the vault keeps working without it. The token stays in the basket and protected for good, and governance can activate it again.
xStocks	Backed's tokenized stocks, such as SPYx and JAAAx, which lose a few base units per transfer.

13.4 Addresses on Ethereum mainnet

This version of the vault is not deployed yet; its address will be added here once it is. The pilot vault deployed on October 27, 2026, at `0x21b823334D2794a637cb9c0fd7fF8671b35F5A23`, runs the previous design, with recorded reserves synced by governance; `docs/SPECIFICATION.md` keeps its record. The components of the pilot basket, in the order of `components()`, with addresses checked against each issuer's official sources:

#	Token	Asset (issuer)	Address
0	ITOTon	iShares Core S&P Total U.S. Stock Market ETF (Ondo)	`0x0692481C369E2BDc728A69ae31b848343a4567Be`
1	SPYx	SPDR S&P 500 ETF Trust (Backed / xStocks)	`0x90A2a4c76b5D8c0bc892A69EA28Aa775a8f2dD48`
2	EFAon	iShares MSCI EAFE ETF (Ondo)	`0x4111b60bc87F2Bd1e81E783E271D7F0ec6EE088B`
3	IEMGon	iShares Core MSCI Emerging Markets ETF (Ondo)	`0xcDD60D15125bf3362b6838D2506b0Fa33bc1a515`
4	SGOVon	iShares 0-3 Month Treasury Bond ETF (Ondo)	`0x8De5D49725550f7b318b2FA0f1B1F118E98E8D0F`
5	BILon	SPDR Bloomberg 1-3 Month T-Bill ETF (Ondo)	`0x2710E30d84B376c34a3B3036aa196b26a83A321b`
6	JAAAx	Janus Henderson AAA CLO ETF (Backed / xStocks)	`0x3bf2E3be4A829Bb3a5f5BE450AE4c8Eb3488da71`
7	PAXG	Physical gold (Paxos)	`0x45804880De22913dAFE09f4980848ECE6EcbAf78`
8	cbBTC	Bitcoin (Coinbase)	`0xcbB7C0000aB88B473b1f5aFd9ef808440eed33Bf`
9	VNQon	Vanguard Real Estate ETF (Ondo)	`0xcCAe8843E26259278C200C6506F6E5A3bdD524cd`
10	DBCon	Invesco DB Commodity Index Tracking Fund (Ondo)	`0x20224080aD516769723c9a4A18325fC4E8C9Ab5D`

13.5 Still open

- **Deployment of this version:** a new vault paired with a guard built for live balances, and any voluntary move from the October 27 pilot vault, which this code cannot change.
- **Name check.** Name and symbol are fixed per contract: a search for other "GRET" tokens on Ethereum and for trademarks on the name. A conflict would need a new vault.
- **Independent audit** of the whole contract before third-party funds.
- **Price coverage.** Six tokens have no Chainlink feed on Ethereum and cannot be swapped; a new guard once feeds appear.
- **An emergency-exit page.** The pilot page has no flow for `redeemSkipping` or for liquidation claims; both work directly on the contract.
- **Governance after the pilot:** the Safe's signers and threshold, whether there will be a timelock and with which delay, and the Governor with VORA.